CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seven Semester of B. Tech. Examination (EC)

December 2012

EC404 Embedded Systems

Date: 06.12.2012, Thursday | **Time: 10:00 a.m. to 01:00 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

Q - 1 (a) Define the following terms: **[03]**
(a)Embedded System, (b)Moore's law, (c)Prototype time

(b) Give an example of a recent consumer product whose prime market window was only about half year. What are its design metrics? Explain each of them. **[04]**

Q - 2 (a) Compare RICS and CICS processor. Why the RISC is more suitable for embedded systems like network processor? **[04]**

(b) For the following C statements, write ARM 7 assembly code: **[04]**

```
int a, b, c, x, y, z;

x = (a+b)-c;

y = a*(b+c) ;

z = (a<<2) | (b&15);
```

(c) Explain common characteristics of embedded system with examples. **[06]**

OR

Q - 2 (a) What are essential structural units in (a) embedded processor, (b) microcontroller? List and explain each of them. **[06]**

(b) List of the branching and arithmetic instructions of SHARC processor. **[04]**

(c) Explain Load-Store architecture of ARM 7 in detail. **[04]**

Q - 3 (a) Explain IC technology in detail. **[05]**

(b) Discuss following content of I2C bus: **[07]**
(a) structure, (b) electrical interface, (c) format of an I2C address transmission, (d)Application interface.

(c) Give the difference between little-endian and big-endian with examples. **[02]**

OR

Q - 3 (a) Explain Application Specific (ASIP) processor. Compare general purpose processor and single purpose processor. **[05]**

(b) Write SHARC assembly code to implement the following C conditional: [05]

```
if ((x-y)>3)
{ a = b - c;
  x = 0; }
else
{ y = 0;
  d = e + f + g ; }
```

(c) Write short note on SHARC link Ports. [04]

SECTION – II

Q - 4 (a) What is an operating system? What are the goals of an operating system design? [05]

(b) What do you mean by 'real time' and 'real time clock'? [02]

Q - 5 (a) Explain process, threads and task in brief. Also describe the characteristics of each. [07]

(b) List out the steps for writing physical device driving ISRs in a system. [07]

OR

Q - 5 (a) What is device driver? Explain virtual devices in brief. [07]

(b) Describe 4 functions and their activities provided by RTOS services. [07]

Q - 6 (a) State and explain any six kernel services. [07]

(b) Explain Design of embedded system by using Automatic Chocolate Vending Machine (ACVM). [07]

OR

Q - 6 (a) What is timer? How does a counter perform (a) timer functions, (b) prefixed time initiated events generation and (c) time capture functions? [06]

(b) What is the role of a scheduler in OS? Describe the 3 model strategies adopted by RTOS Services. [08]

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